Biological molecules

Mark scheme and worked solutions · 20 questions

Review export of the existing pipeline-approved batch. Final owner review pending.

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Question 1

core · 1 mark · 0610_T04_P1_20261013_Q1

Answer: B

Rationale
Carbohydrates and fats contain only carbon, hydrogen and oxygen. Proteins contain these three elements and also nitrogen. Carbon, oxygen and hydrogen are wrong because they are found in all three types of molecule.

Question 2

core · 1 mark · 0610_T04_P1_20261013_Q2

Answer: D

Rationale
Starch is made from glucose. Cellulose is made from glucose, not amino acids. Glycogen is made from glucose, not fatty acids and glycerol. Proteins are made from amino acids, not glucose.

Question 3

core · 1 mark · 0610_T04_P1_20261013_Q3

Answer: A

Rationale
Only the biuret test was positive (purple), so protein is present. Iodine staying orange-brown means no starch, so the options with starch are wrong. Benedict's staying blue after heating means no reducing sugar, so "protein and reducing sugar" is wrong.

Question 4

core · 1 mark · 0610_T04_P1_20261013_Q4

Answer: C

Rationale
In the ethanol emulsion test the sample is shaken with ethanol and the ethanol is poured into water; a milky white emulsion shows fat or oil. Orange-brown is the colour of iodine solution itself, so it is a negative result for starch (positive is blue-black). Benedict's solution must be heated to give a result. DCPIP starts blue and turns colourless with vitamin C, not the other way round.

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Question 5

core · 1 mark · 0610_T04_P1_20261013_Q5

Answer: A

Rationale
The more vitamin C a drink contains, the fewer drops are needed to decolourise the same volume of DCPIP. Drink 2 needed the fewest drops (6), so it has the highest concentration. Drink 3 needed the most drops, so it has the lowest concentration, not the highest.

Question 6

core · 1 mark · 0610_T04_P1_20261013_Q6

Answer: B

Rationale
Starch, glycogen and cellulose are all large carbohydrate molecules made from glucose. Proteins, not these carbohydrates, are made from amino acids and contain nitrogen. All carbohydrates contain carbon, hydrogen and oxygen, so it is wrong that only starch contains carbon.

Question 7

core · 1 mark · 0610_T04_P1_20261013_Q7

Answer: D

Rationale
With Benedict's solution, the colour changes from blue through green, yellow and orange to brick red as the concentration of reducing sugar increases. So the order is blue, green, yellow, brick red. The first option is the order of decreasing concentration. The other two put green and yellow in the wrong places in the colour sequence.

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Question 8

supplement · 1 mark · 0610_T04_P1_20261013_Q8

Answer: C

Rationale
DNA has two strands coiled together to form a double helix, and bonds between pairs of bases hold the strands together. DNA has two strands, not one. The bonds holding the strands together are between bases, not glucose molecules. Base pairing is always A with T and C with G, not A with C.

Question 9

supplement · 1 mark · 0610_T04_P1_20261013_Q9

Answer: B

Rationale
Base C pairs with base G, so G is also 24%. C and G together make 48%, leaving 52% for A and T. A pairs with T, so each is half of 52%, which is 26%. 24% wrongly assumes A equals C; 48% is C plus G; 52% is A plus T together.

Worked solution
Percentage of G = percentage of C = 24%. Percentage of A + T = 100 − 24 − 24 = 52%. Percentage of A = 52 ÷ 2 = 26%. Check: 26 + 26 + 24 + 24 = 100.

Question 10

core · 1 mark · 0610_T04_P1_20261013_Q10

Answer: A

Rationale
Of carbohydrates, fats and proteins, only proteins contain nitrogen, so the powder is a protein and gives a purple colour with biuret solution. The iodine test detects starch, the ethanol emulsion test detects fats, and the Benedict's test detects reducing sugars; carbohydrates and fats contain only carbon, hydrogen and oxygen.

Question 11

core · 5 marks · 0610_T04_P1_20261013_Q11

(a) [1]

- carbon, hydrogen and oxygen (all three needed) (1)
- ALLOW: C, H, O
- REJECT: nitrogen included in the list

(b) [1]

- glycogen / cellulose (1)

(c) [1]

- amino acids (1)

(d) [2]

- fatty acids (1)
- glycerol (1)

Worked solution
Carbohydrates and fats are both made of only three elements: carbon, hydrogen and oxygen. Proteins contain these three and nitrogen as well. Glucose is the building block for three large carbohydrates: starch, glycogen and cellulose, so either glycogen or cellulose answers (b). Proteins are made from many amino acids joined together. Fats (and oils) are made from two kinds of smaller molecule: fatty acids and glycerol.

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Question 12

core · 4 marks · 0610_T04_P1_20261013_Q12

(a) [1]

- iodine solution (1)
- ALLOW: iodine
- REJECT: Benedict's solution / biuret solution

(b) [1]

- blue-black (1)
- ALLOW: black / dark blue

(c) [1]

- orange-brown (1)
- ALLOW: orange / brown / yellow-brown / stays the same colour as the iodine
- REJECT: colourless

(d) [1]

- glucose (1)

Worked solution
Iodine solution is the test for starch. Iodine solution is orange-brown; if starch is present the colour changes to blue-black. If there is no starch the mixture stays orange-brown, the colour of the iodine solution itself. Starch is a large carbohydrate molecule made of many glucose molecules joined together.

Question 13

core · 6 marks · 0610_T04_P1_20261013_Q13

(a) [2]

- add ethanol to the crushed peanuts and shake (1)
- pour the ethanol (liquid) into water in a (clean) test-tube (1)
- ALLOW: add water to the ethanol / mix the ethanol with water
- IGNORE: references to safety or to filtering

(b) [1]

- milky white emulsion forms (1)
- ALLOW: cloudy (white) / white emulsion
- REJECT: colour changes to purple / blue-black

(c) [1]

- as a control, to compare with / to show that a milky result is caused by the peanuts and not by the ethanol or water (1)
- ALLOW: to show what a negative result looks like

(d) [2]

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- add biuret solution to (a sample of) the peanuts (mixed with water) (1)
- ALLOW: add sodium hydroxide solution then (dilute) copper sulfate solution
- purple colour shows protein (1)
- ALLOW: violet / lilac / mauve
- REJECT: pink / red

Worked solution
Fats dissolve in ethanol but not in water. In the ethanol emulsion test the crushed food is shaken with ethanol so that any fat dissolves. The ethanol is then poured into water. If fat is present it comes out of solution as tiny droplets, making a milky white emulsion. Testing distilled water in the same way is a control: it shows what happens when no fat is present, so Kofi can be sure a milky result is due to the peanuts. For protein, biuret solution is added; it is blue, and turns purple if protein is present.

Question 14

core · 5 marks · 0610_T04_P1_20261013_Q14

(a) [1]

- blue to colourless (1)
- REJECT: colourless to blue / blue to pink

(b) [1]

- 2.7 (cm^3) (1)

(c) [2]

- fresh juice (1)
- (fresh juice) needed a smaller (mean) volume (1.3 cm^3 compared with 2.7 cm^3) to decolourise the same volume of DCPIP (1)
- ECF: from the mean calculated in (b)
- REJECT: fresh juice needed more juice

(d) [1]

- any one from: volume of DCPIP / concentration of DCPIP / same oranges (or same carton) for both juices / temperature of the juice when tested / same way of adding juice (slowly, with mixing) (1)
- IGNORE: amount of juice (this is what is measured)

Worked solution
DCPIP is a blue solution that becomes colourless when vitamin C is added. The mean for boiled juice is (2.5 + 2.7 + 2.9) ÷ 3 = 8.1 ÷ 3 = 2.7 cm^3. Check: 2.7 is in the middle of the three values, which are evenly spaced. The fresh juice needed only 1.3 cm^3 to decolourise the same 1.0 cm^3 of DCPIP, so each cm^3 of fresh juice contains more vitamin C. The less juice needed, the more vitamin C it contains. To make it a fair comparison, everything else, such as the volume and concentration of DCPIP, must stay the same.

Question 15

supplement · 4 marks · 0610_T04_P1_20261013_Q15

(a) [3]

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- any three from:
- two strands (1)
- (strands) coiled (together) to form a double helix (1)
- each strand contains bases (1)
- bonds between (pairs of) bases hold the strands together (1)
- A pairs with T and C pairs with G (1)
- ALLOW: hydrogen bonds between bases
- IGNORE: full names of bases

(b) [1]

- base A always pairs with base T (1)
- ALLOW: A pairs with T and C pairs with G

Worked solution
DNA is made of two strands that coil around each other to form a double helix. Each strand contains chemicals called bases. Bonds between pairs of bases, one on each strand, hold the two strands together. The bases always pair in the same way: A with T, and C with G. So the student's sentence is wrong: A pairs with T, not with G.

Question 16

supplement · 4 marks · 0610_T04_P1_20261013_Q16

(a) [2]

- 27(%) (1)
- A (always) pairs with T, so there is the same amount of each (1)

(b) [2]

- 100 − (27 + 27) = 46 (%), shared equally between C and G (1)
- 23 (%) (1)
- ECF: from the value of T in (a)

Worked solution
In DNA, every A on one strand is paired with a T on the other strand, so the percentage of T equals the percentage of A: 27%. A and T together make 27 + 27 = 54%. The rest, 100 − 54 = 46%, is C and G. Every C pairs with a G, so they are equal: 46 ÷ 2 = 23% G. Check: 27 + 27 + 23 + 23 = 100%.

Question 17

core · 6 marks · 0610_T04_P1_20261013_Q17

(a) [4]

- J: protein (solution) (1)
- K: starch (suspension) (1)
- L: glucose (solution) (1)
- M: distilled water (1)

(b) [2]

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- starch (1)
- reducing sugar (1)
- ALLOW: glucose for reducing sugar
- REJECT: protein (each wrong extra nutrient cancels one mark)

Worked solution
Look at which test is positive for each tube. J gives purple with biuret only, so it is protein. K gives blue-black with iodine only, so it is starch. L gives brick red with Benedict's only, so it is glucose, a reducing sugar. M is negative for all three tests (iodine stays orange-brown, Benedict's and biuret stay blue), so it is distilled water. Liquid N is positive for iodine (blue-black) and positive for Benedict's (orange means some reducing sugar), but negative for biuret, so it contains starch and reducing sugar but no protein.

Question 18

core · 6 marks · 0610_T04_P1_20261013_Q18

(a) [2]

- blue-black at the start shows starch present, orange-brown after 30 minutes shows no starch (left) (1)
- the starch has been broken down (into smaller molecules / sugar) by (a substance in) the barley liquid (1)
- ALLOW: broken down by an enzyme / amylase
- ALLOW: digested / hydrolysed for broken down

(b) [2]

- no reducing sugar at the start, but reducing sugar present after 30 minutes (1)
- starch is made from glucose, so breaking down starch produces (reducing) sugar (1)
- ALLOW: glucose / maltose for reducing sugar
- ALLOW: enzyme / amylase (in the barley liquid) breaks down the starch

(c) [1]

- the liquid does not break down cellulose (into sugar) (1)
- ALLOW: the liquid only breaks down starch / the enzyme is specific to starch / the liquid has no enzyme for cellulose

(d) [1]

- as a control, to show the seed liquid itself contains no reducing sugar, so the sugar came from the starch (1)

Worked solution
At the start, iodine turns blue-black, so starch is present. After 30 minutes the iodine stays orange-brown, so the starch has gone: the barley liquid has broken it down. Benedict's stays blue at the start, so there is no reducing sugar. After 30 minutes it turns orange, so reducing sugar is now present. Starch is made of glucose units joined together, so breaking it down releases sugar. Cellulose is also made from glucose, but no sugar appeared, so the liquid cannot break down cellulose. The test on the barley liquid alone shows it contained no reducing sugar of its own, so the sugar must have come from the starch.

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Question 19

supplement · 6 marks · 0610_T04_P1_20261013_Q19

(a) [1]

- G C T A A T G C C T (1)

(b) [2]

- 5 C bases (2 on strand 1 + 3 on strand 2) (1)
- (5 ÷ 20) × 100 = 25 (%) (1)
- ECF: from the strand 2 sequence in (a)

(c) [2]

- bonds between (pairs of) bases (1)
- because these bonds hold the two strands together (1)
- ALLOW: hydrogen bonds

(d) [1]

- every A on one strand pairs with a T on the other strand (1)

Worked solution
Each base on strand 2 pairs with the base opposite it on strand 1: A with T, T with A, C with G and G with C. So strand 2 reads G C T A A T G C C T. Strand 1 has 2 C bases and strand 2 has 3 C bases, making 5 out of 20: 5 ÷ 20 × 100 = 25%. Check: strand 1 has 3 G bases, which pair with the 3 C bases on strand 2, and the total number of G bases (5) equals the total number of C bases (5). Heating separates the two strands but leaves each strand whole, so the broken bonds must be the ones that hold the strands together: the bonds between pairs of bases. The number of A bases equals the number of T bases because every A is paired with a T on the opposite strand.

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Question 20

core · 6 marks · 0610_T04_P1_20261013_Q20

(a) [2]

- add (the same volume of) Benedict's solution to the extract (1)
- heat in a (hot / boiling) water bath for a few minutes (1)
- ALLOW: heat to about 80 °C
- REJECT: warm in the hand / leave at room temperature

(b) [2]

- starch present in unripe and ripe bananas but absent in overripe bananas / starch decreases (1)
- reducing sugar increases as the banana ripens (green, then orange, then brick red) (1)

(c) [2]

- any two from: volume of extract tested / volume of Benedict's solution / temperature of the water bath / heating time / volume of water used to make the extract (2)

Worked solution

For the Benedict's test, add Benedict's solution to the extract and heat it in a hot water bath; without heating there is no colour change. As the banana ripens, the Benedict's colour changes from green to orange to brick red, so the reducing sugar content increases. The iodine test stays blue-black until the banana is overripe, when it stays orange-brown, so the starch content falls and is finally all gone. For the Benedict's colours to be compared fairly, each test must use the same volume of extract and of Benedict's solution and must be heated at the same temperature for the same time.

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